A Data sources and cohorts

MMRF CoMMpass, interim analysis 24 (IA24)

MMRF CoMMpass (IA24)
1,143 subject records

Clinical data
diagnosis · laboratory · treatment · response · adverse events · symptoms

Bulk RNA sequencing
TPM → ssGSEA (MSigDB Hallmark): 50 pathway scores per RNA sample

Visit pairs V(t) → V(t+1)
13,451 pairs
1,025 patients with ≥1 pair

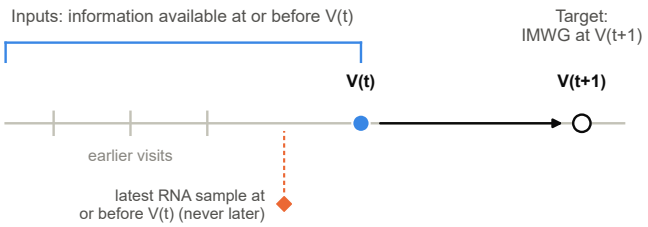
RNA available at V(t)
9,173 pairs · 707 patients
(68% of pairs)

Six-class response task
13,451 pairs · 1,025 patients
IMWG: PD/SD/PR/VGPR/CR/sCR

Binary improvement task
12,298 pairs · 1,018 patients
excludes 1,153 pairs with sCR at V(t)

Survival cohort: 674 patients
RNA by day 0 · PFS 459 events · OS 247 deaths

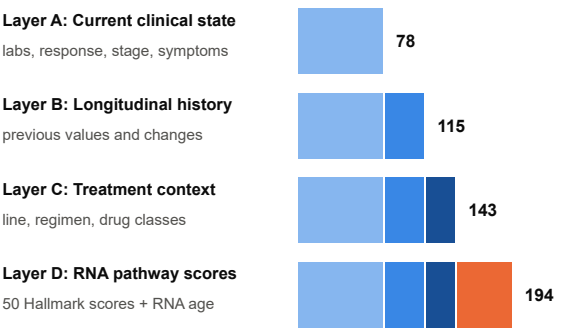
B Visit-pair construction



Six-class: PD, SD, PR, VGPR, CR or sCR at V(t+1).
Binary: V(t+1) is a better IMWG category than V(t).
Every pair of a patient stays in one data partition.

C Information layers A–D

Nested feature sets, not algorithms

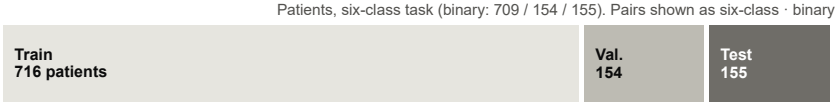


Bars: cumulative source input features, coloured by the layer that adds them (D adds 50 pathway scores + RNA age, plus a derived RNA-availability indicator).

D Patient-level partitions and modelling framework

Assigned once; shared by every layer and algorithm; no patient is in more than one partition

Patients, six-class task (binary: 709 / 154 / 155). Pairs shown as six-class · binary



Train: tuning and fitting
Preprocessing fitted; hyperparameters chosen by 5-fold patient-grouped CV on these patients only
Pairs: six-class 9,539 · binary 8,709

Validation
Binary decision threshold; calibration comparison
Pairs: 1,833 · 1,659

Test
Scored once; patient-level bootstrap CIs
Pairs: 2,079 · 1,930

Modelling framework

- ● Logistic Regression
- ■ LightGBM
- ◆ XGBoost

4 layers × 3 algorithms × 2 tasks = 24 models

Binary: AUPRC (test improvement rate 18.3%)
Six-class: macro-F1 (argmax)
Also AUROC, Brier score, RPS, log loss

E Evaluation, downstream analyses and validation status

All evaluations are internal to CoMMpass; no external validation cohort was used

Analysis	Patients and design	Validation status
Main prediction (24 models), calibration and SHAP	Test: 155 patients, scored once. Tuning on train patients only	Internal held-out test split; calibration and SHAP descriptive
RNA contribution, uncertainty and subgroup analyses	5×5 grouped CV on train + validation patients; train-tuned hyperparameters	Post hoc, exploratory; CV cohort overlaps the tuning patients
Proliferation / MYC association	Discovery: train + validation. Test: 84 patients with RNA	**Not independent; redesign pending** Pathway family chosen using pooled data that included test patients: not independent
Survival (PFS, OS) and pathway coefficients	674 patients, 5×5 CV; includes 101 validation and 100 test patients	Exploratory internal CV; overlaps other partitions
Geographic hold-out (US → Spain, Canada)	Fit: 632 US patients; threshold and calibration: 139 US patients. Scored once: 116 Spain/Canada patients	**Not independent; redesign pending** 80 of 116 evaluated patients were in the training split used to tune hyperparameters

Amber rows: known validation concerns under revision. Hyperparameters were tuned on train patients only in the main models; the scope of excluding validation patients from all development steps (tuning, pathway selection) is being clarified. Nothing here is externally validated, and CIs are descriptive and unadjusted.